

# **Residential Lighting System Design**

**Engineering Design Project**

**Evin Nisa Terat**

**Electrical and Electronics Engineering  
Muğla Sıtkı Koçman University**

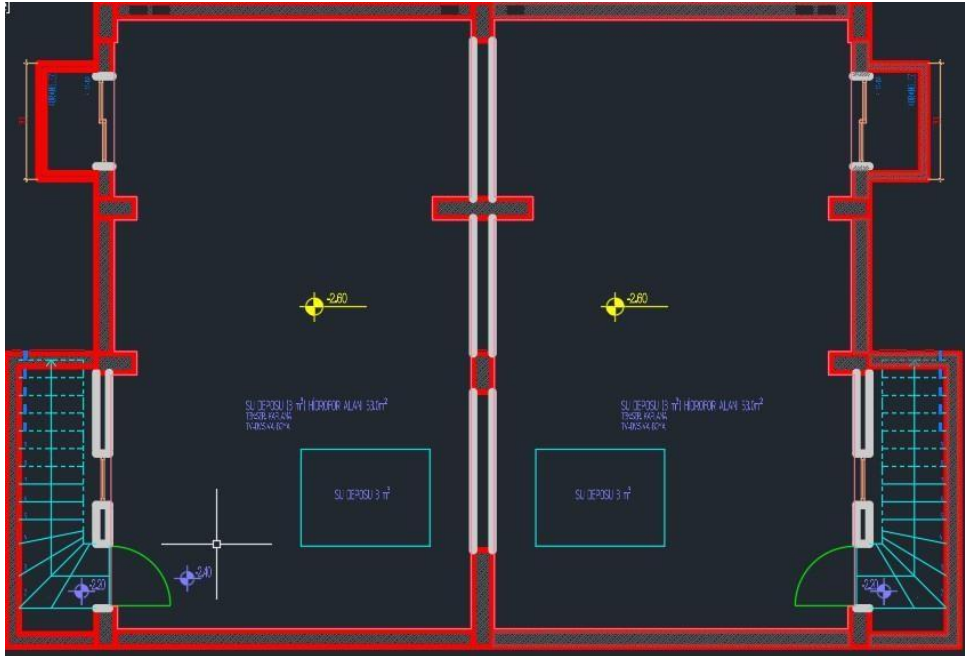
**May 2026**

# CONTENTS

|                                                                                  |    |
|----------------------------------------------------------------------------------|----|
| CONTENTS .....                                                                   | 1  |
| Architectural Plan: .....                                                        | 2  |
| FIRST HOUSE .....                                                                | 2  |
| Luminous Flux and Power Calculations for Internal and External Illumination..... | 2  |
| Light Sources and Luminaires .....                                               | 3  |
| Loading Table .....                                                              | 4  |
| Cost Of Project.....                                                             | 6  |
| SECOND HOUSE .....                                                               | 6  |
| Luminous Flux and Power Calculations for Internal and External Illumination..... | 7  |
| Light Sources and Luminaires .....                                               | 12 |
| Loading Table .....                                                              | 12 |
| Cost Of Project.....                                                             | 14 |
| Video of Project .....                                                           | 14 |
| <a href="https://youtu.be/eYsRW0s9ccM">https://youtu.be/eYsRW0s9ccM</a> .....    | 14 |

## Architectural Plan:

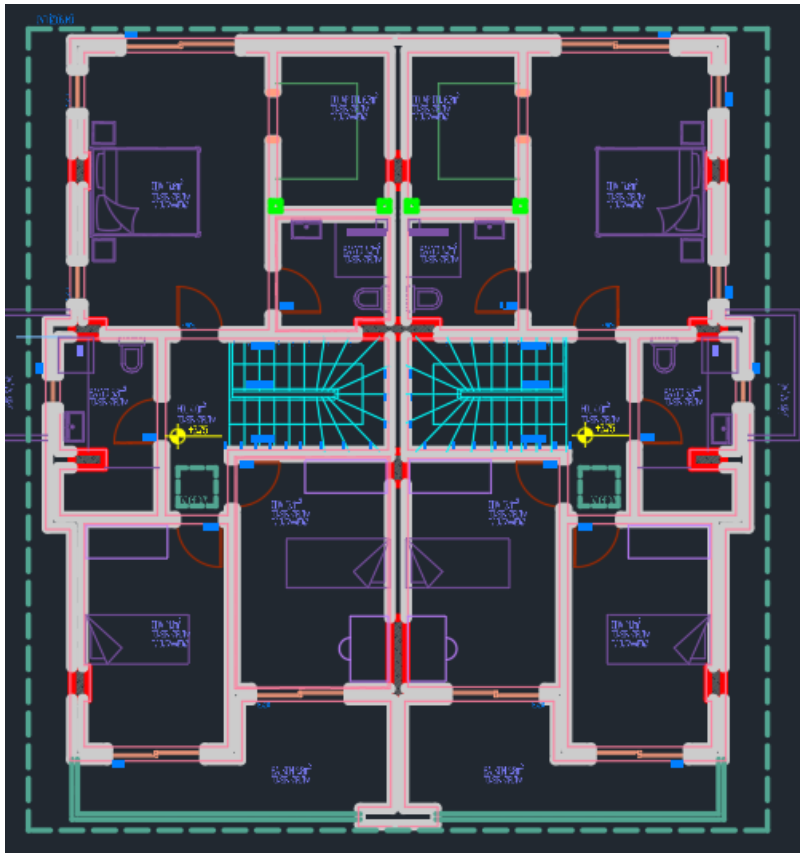
Basement:



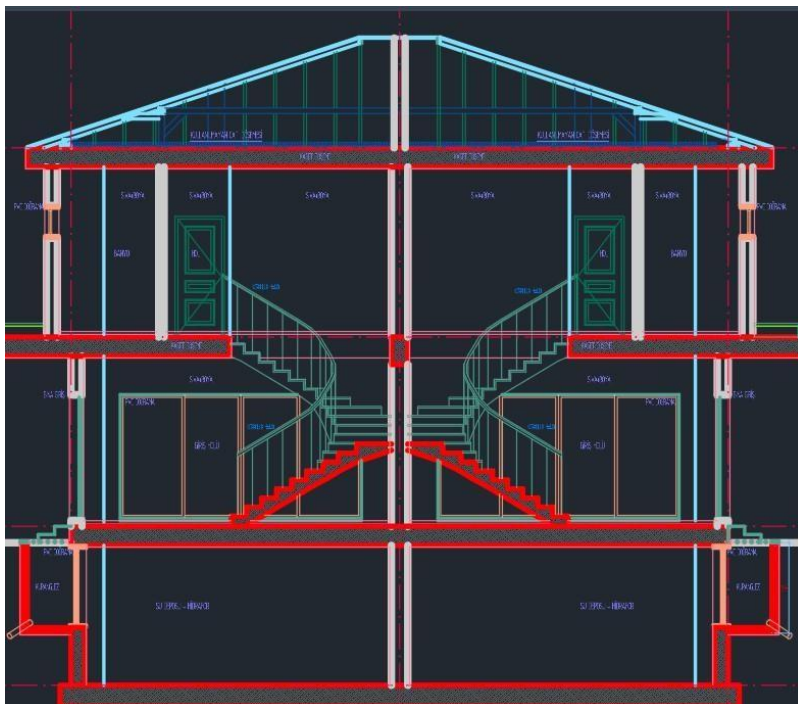
Ground Floor:



Upstair:



Side:



## FIRST HOUSE

### Luminous Flux and Power Calculations for Internal and External Illumination

Basement:

$$A = 53.0 \text{ m}^2 \text{ target lux: } 50 \quad b = 5.8 \text{ m} \quad l = 9.14 \text{ m} \quad h_c = 2.80 - (0.0 + 0.10) = 2.7 \text{ m}$$

$$k = \frac{8b + 2l}{10h_c} \quad \Phi_{cal} = \frac{E \cdot A \cdot d}{\eta}$$

$$k = \frac{(5.8) \times 8 + (9.14) \times 2}{10 \times 2.7} = 2.4 \quad \eta = 0.39$$

$$d = 1.45$$

$$\Phi_{cal} = \frac{53.0 \times 50 \times 1.45}{0.39} = 9850 \text{ lm} \quad (\text{Incandescent spherical diffuser})$$

for 150w incandescent lamp (2200 lm)

$$n = \frac{9850}{2200} = 4.47 \rightarrow 4 \text{ units}$$

Living Room:

$$A = 24.5 \text{ m}^2 \text{ target lux: } 100 \quad b = 4.0 \text{ m} \quad l = 6.12 \text{ m} \quad h_c = 2.80 - (0.30 + 0.75) = 1.75 \text{ m}$$

$$k = \frac{(4.0) \times 8 + (6.12) \times 2}{10 \times 1.75} = 2.5 \quad \eta = 0.39$$

$$d = 1.45$$

$$\Phi_{cal} = \frac{24.5 \times 100 \times 1.45}{0.39} = 9110 \text{ lm} \quad (\text{Incandescent spherical diffuser})$$

for 100w incandescent lamp (1400 lm)

$$n = \frac{9110}{1400} = 6.5 \rightarrow 6 \text{ unit} \rightarrow 2 \text{ units } 3 \times 100 \text{ w}$$

Entrance Hall:

$$A = 5.7 \text{ m}^2 \text{ target lux: } 100 \quad b = 2.1 \text{ m} \quad l = 2.7 \text{ m} \quad h_c = 2.80 - (0.20 + 0.0) = 2.60 \text{ m}$$

$$k = \frac{2.1 \times 8 + 2.7 \times 2}{10 \times 2.6} = 0.85 \quad \eta = 0.24$$

$$d = 1.55$$

$$\Phi_{cal} = \frac{5.7 \times 100 \times 1.55}{0.24} = 3680 \text{ lm} \quad (\text{Plastic diffuser LED direct lighting})$$

for 18w LED panel (1800 lm)

$$n = \frac{3680}{1800} = 2.04 \rightarrow 2 \text{ unit}$$

Stairs 2:

$$A = 4.25 \text{ m}^2 \text{ target lux: } 50 \quad b = 1.0 \text{ m} \quad l = 4.25 \text{ m} \quad h_c = 2.80 - 0.10 = 2.70 \text{ m}$$

$$k = \frac{1 \times 8 + 4.25 \times 2}{10 \times 2.7} = 0.6 \quad \eta = 0.29 \quad d = 1.45$$

$$\Phi_{cal} = \frac{4.25 \times 50 \times 1.45}{0.29} = 1062 \text{ lm} \quad (\text{Semi-indirect lighting with translucent Reflector})$$

for Iph44 glass globe bow incandescent (800 lm)

$$n = \frac{1062}{800} = 1.3 \rightarrow 1 \text{ unit}$$



### Stairs:

$$A = 6.0 \text{ m}^2 \text{ target lux: } 150 \quad b = 2.2 \text{ m} \quad l = 2.75 \text{ m} \quad h_k = 2.80 - 0.20 = 2.60$$

$$k = \frac{2.2 \times 8 + 2.75 \times 2}{10 \times 2.6} = 0.83 \quad \eta = 0.24$$

$$d = 1.55$$

$$\Phi_{cal} = \frac{6.0 \times 150 \times 1.55}{0.24} = 5812.5 \text{ lm} \quad (\text{Plastic diffuser LED direct lighting})$$

for 16w LED panel

$$n = \frac{5812.5}{1500} = 3.87 \rightarrow 4 \text{ unit}$$

### Lower Hall:

$$A = 2.0 \text{ m}^2 \text{ target lux: } 100 \quad b = 1.0 \text{ m} \quad l = 2.0 \text{ m} \quad h_k = 2.80 - 0.20 = 2.60$$

$$k = \frac{1.0 \times 8 + 2.0 \times 2}{10 \times 2.6} = 0.46$$

$$\eta = 0.24$$

$$d = 1.55$$

$$\Phi_{cal} = \frac{2.0 \times 100 \times 1.55}{0.24} = 1300 \text{ lm} \quad (\text{Plastic diffuser LED direct lighting})$$

for 15w LED Globe (1300lm)

$$n = \frac{1300}{1300} = 1 \text{ unit}$$

### Restroom:

$$A = 2.3 \text{ m}^2 \text{ target lux: } 100 \quad b = 1.3 \text{ m} \quad l = 1.77 \text{ m} \quad h_k = 2.80 - (0.05 + 0.85) = 1.90 \text{ m}$$

$$k = \frac{1.3 \times 8 + 1.77 \times 2}{10 \times 1.9} = 0.73$$

$$\eta = 0.20$$

$$d = 1.55$$

$$\Phi_{cal} = \frac{2.3 \times 100 \times 1.55}{0.20} = 1700 \text{ lm} \quad (\text{Semi-indirect Lighting with Plastic diffuser})$$

for 18w IP44 LED (1800lm)

$$n = \frac{1700}{1800} = 0.94 \rightarrow 1 \text{ unit}$$

### Kitchen:

$$A = 12.2 \text{ m}^2 \text{ target lux: } 300 \quad b = 3.0 \text{ m} \quad l = 4.07 \text{ m} \quad h_k = 2.80 - (0.30 + 0.75) = 1.75 \text{ m}$$

$$k = \frac{3 \times 8 + 4.07 \times 2}{10 \times 1.75} = 1.9$$

$$\eta = 0.40$$

$$d = 1.55$$

$$\Phi_{cal} = \frac{12.2 \times 300 \times 1.55}{0.40} = 14100 \text{ lm} \quad (\text{Plastic diffuser LED direct lighting})$$

for 60x60 cm LED panel (4000lm)

$$n = \frac{14100}{4000} = 3.5 \text{ units} \rightarrow 2 \text{ units}$$

"As the kitchen area is under  $20 \text{ m}^2$ , using 2 panels instead of 4 is more appropriate to provide a balanced illumination level without causing visual fatigue."



Upper hall:

$$A = 4.0 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 1.8 \text{ m} \quad l = 2.22 \text{ m} \quad h_c = 2.80 - 0.20 = 2.60$$

$$k = \frac{1.8 \times 8 + 2.22 \times 2}{10 \times 2.60} = 0.72 \quad \eta = 0.24 \quad d = 1.55$$

$$\Phi_{cal} = \frac{4.0 \times 100 \times 1.55}{0.24} = 2580 \text{ lm} \quad (\text{Plastic diffuser LED direct lighting})$$

for 15W LED Globe (1300 lm)

$$n = \frac{2580}{1300} \approx 2 \text{ units} \rightarrow 2 \text{ units}$$

Master Bedroom:

$$A = 17.6 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 3.8 \text{ m} \quad l = 4.63 \text{ m} \quad h_c = 2.80 - (0.30 + 0.75) = 1.75 \text{ m}$$

$$k = \frac{3.8 \times 8 + 4.63 \times 2}{10 \times 1.75} = 2.27 \quad \eta = 0.39 \quad d = 1.45$$

$$\Phi_{cal} = \frac{17.6 \times 100 \times 1.45}{0.39} = 6540 \text{ lm} \quad (\text{Incandescent spherical diffuser})$$

for 100W incandescent lamp (1400 lm)

$$n = \frac{6540}{1400} = 4.7 \rightarrow 4 \text{ units} \rightarrow 2 \text{ unit } 2 \times 100 \text{ W}$$

Closet room:

$$A = 6.2 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 2 \text{ m} \quad l = 3.1 \text{ m} \quad h_c = 2.80 - 0.85 = 1.95 \text{ m}$$

$$k = \frac{2 \times 8 + 3.1 \times 2}{10 \times 1.95} = 1.14 \quad \eta = 0.28 \quad d = 1.55$$

$$\Phi_{cal} = \frac{6.2 \times 100 \times 1.55}{0.28} = 3430 \text{ lm} \quad (\text{Plastic diffuser LED direct lighting})$$

for 18W LED panel (1800 lm)

$$n = \frac{3430}{1800} = 1.9 \rightarrow 2 \text{ units}$$

Master Bathroom:

$$A = 4.2 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 1.8 \text{ m} \quad l = 2.33 \text{ m} \quad h_c = 2.80 - 0.85 = 1.95$$

$$k = \frac{1.8 \times 8 + 2.33 \times 2}{10 \times 1.95} = 0.98 \quad \eta = 0.29 \quad d = 1.45$$

$$\Phi_{cal} = \frac{4.2 \times 100 \times 1.45}{0.29} = 2100 \text{ lm} \quad (\text{Semi-indirect lighting with Plastic diffuser})$$

for 100W IP44 glass globe incandescent (1400 lm)

$$n = \frac{2100}{1400} = 1.5 \rightarrow 1 \text{ unit}$$

and  
for 10W LED  $\Rightarrow$  Mirner (1000 lm)

$$n = \frac{700}{1000} = 0.7 \rightarrow 1 \text{ unit}$$



### Balcony:

$$A = 9,8, 0 \text{ m}^2 \text{ target lux: } 50 \quad b = 2,2 \text{ m} \quad l = 4,45 \text{ m} \quad h_L = 2,80 - 0,10 = 2,70$$

$$k = \frac{2,2 \times 8 + 4,45 \times 2}{10 \times 2,70} = 0,9 \quad \eta = 0,29 \quad d = 1,45$$

$$\Phi_{cal} = \frac{9,8 \times 50 \times 1,45}{0,29} = 2450 \text{ lm} \quad (\text{Semi-indirect lighting with translucent Reflector})$$

for IP44 glass globe 100w incandescent (1400 lm)

$$n = \frac{2450}{1400} = 1,75 \rightarrow 2 \text{ units}$$

### Room 1:

$$A = 12,1 \text{ m}^2 \text{ target lux: } 75 \quad b = 3 \text{ m} \quad l = 4,03 \text{ m} \quad h_L = 2,80 - (0,25 + 0,85) = 1,70 \text{ m}$$

$$k = \frac{3 \times 8 + 4,03 \times 2}{10 \times 1,70} = 1,88 \quad \eta = 0,34 \quad d = 1,45$$

$$\Phi_{cal} = \frac{12,1 \times 75 \times 1,45}{0,34} = 3800 \text{ lm} \quad (\text{Incandescent spherical diffuser})$$

for 100 w incandescent lamp (1400 lm)

$$n = \frac{3800}{1400} = 2,71 \rightarrow 3 \text{ unit} \rightarrow 1 \text{ unit } 2 \times 100 \text{ w}$$

### Room 2:

$$A = 11,1 \text{ m}^2 \text{ target lux: } 75 \quad b = 3 \text{ m} \quad l = 3,7 \text{ m} \quad h_L = 2,80 - (0,30 + 0,75) = 1,75$$

$$k = \frac{3 \times 8 + 3,7 \times 2}{10 \times 1,75} = 1,8 \quad \eta = 0,34 \quad d = 1,45$$

$$\Phi_{cal} = \frac{11,1 \times 75 \times 1,45}{0,34} = 3550 \text{ lm} \quad (\text{Incandescent spherical diffuser})$$

for 100w incandescent lamp (1400 lm)

$$n = \frac{3550}{1400} = 2,54 \rightarrow 3 \text{ unit} \rightarrow 1 \text{ unit } 2 \times 100 \text{ w}$$

### Bathroom (1st floor):

$$A = 5,8 \text{ m}^2 \text{ target lux: } 100 \quad b = 2,1 \text{ m} \quad l = 2,76 \text{ m} \quad h_L = 2,80 - (0,85) = 1,95 \text{ m}$$

$$k = \frac{2,1 \times 8 + 2,76 \times 2}{10 \times 1,95} = 1,1 \quad \eta = 0,29 \quad d = 1,45$$

$$\Phi_{cal} = \frac{5,8 \times 100 \times 1,45}{0,29} = 2800 \text{ lm} \quad (\text{Semi-indirect lighting with translucent Reflector})$$

for 100w IP44 glass globe incandescent (1400 lm)

$$n = \frac{2800}{1400} = 2 \rightarrow 1 \text{ unit}$$

2nd  
for 10w LED for the mirror (1000 lm)



| Room / Area          | Area (m²) | Target Lux (lx) | Calculated Lumens (lm) | Fixture / Lamp Selection                     |
|----------------------|-----------|-----------------|------------------------|----------------------------------------------|
| Living Room          | 24.5      | 100             | 9110                   | 2 units 3x100W incandescent lamp             |
| Entrance Hall        | 5.7       | 100             | 3680                   | 2 units 18W LED panel                        |
| Stairs               | 6.0       | 150             | 5812.5                 | 4 Units 16W Led panel                        |
| Lower Hall           | 2.0       | 100             | 1300                   | 15W LED Globe                                |
| Restroom             | 2.3       | 100             | 1700                   | 18W IP44 LED                                 |
| Kitchen              | 12.2      | 300             | 14182                  | 2 units 60x60 LED panel                      |
| Basement             | 53.0      | 50              | 9852                   | 4 units 150W incandescent lamp               |
| Balcony              | 9.8       | 50              | 2450                   | 2 unit IP44 Glass Globe<br>100W Incandescent |
| Room 1               | 12.1      | 75              | 3800                   | 2x100W incandescent lamp                     |
| Room 2               | 11.1      | 75              | 3550                   | 2x100W incandescent lamp                     |
| Bathroom (1st Floor) | 5.8       | 100             | 2800                   | 100W IP44 Glass Globe incandescent& 10W LED  |
| Upper Hall           | 4.0       | 100             | 2583                   | 2 units 15W LED Globe                        |
| Master Bedroom       | 17.6      | 100             | 6543                   | 2 units 2x100W incandescent lamp             |
| Closet Room          | 6.2       | 100             | 3472                   | 2 units 18W LED panel                        |
| Master Bathroom      | 4.2       | 100             | 2100                   | 100W & 10W LED                               |
| Stairs 2             | 4.25      | 50              | 1062                   | 60W IP44 Glass Globe incandescent            |

## Light Sources and Luminaires

| Code | Luminaire Type              | Lamp Type            | Power (W) | Luminous Flux (lm) | IP Rating |
|------|-----------------------------|----------------------|-----------|--------------------|-----------|
| L1   | Ceiling Mounted Luminaire   | Incandescent Lamp    | 100 W     | ~1300 lm           | IP20      |
| L2   | Ceiling Mounted Luminaire   | 2x Incandescent Lamp | 2x100 W   | ~2600 lm           | IP20      |
| P1   | LED Panel (60x60)           | Integrated LED       | 36 W      | ~3400 lm           | IP20      |
| P2   | LED Panel / Surface Mounted | Integrated LED       | 18 W      | ~1600 lm           | IP20      |
| P3   | LED Panel / Surface Mounted | Integrated LED       | 16 W      | ~1400 lm           | IP20      |
| P4   | LED Panel / Surface Mounted | Integrated LED       | 12 W      | ~1000 lm           | IP20      |
| G1   | Waterproof Glass Globe      | Incandescent Lamp    | 100 W     | ~1300 lm           | IP44      |
| S1   | Recessed LED Spot           | Integrated LED       | 9 W       | ~800 lm            | IP20      |



**Loading Table**

| <b>Line No</b> | <b>Power (W)</b> | <b>Load Type</b> | <b>Outlet/Light Qty</b> | <b>Current (A)</b> | <b>Section (mm2)</b> | <b>Fuse (A)</b> | <b>Distance (m)</b> | <b>Volt. Drop (e%)</b> |
|----------------|------------------|------------------|-------------------------|--------------------|----------------------|-----------------|---------------------|------------------------|
| <b>1</b>       | 2000             | Oven             | 1                       | 9.09               | 2.5                  | 16              | 7.89                | 0.04                   |
| <b>2</b>       | 2000             | Kettle           | 1                       | 9.09               | 2.5                  | 16              | 6.44                | 0.03                   |
| <b>3</b>       | 2500             | Dishwasher       | 1                       | 11.36              | 2.5                  | 16              | 4.36                | 0.03                   |
| <b>4</b>       | 2200             | Mixed Outlets    | 6                       | 10.00              | 2.5                  | 16              | 14.84               | 0.09                   |
| <b>5</b>       | 1500             | TV + Sockets     | 5                       | 6.82               | 2.5                  | 16              | 16.24               | 0.07                   |
| <b>6</b>       | 600              | Lighting         | 4                       | 2.73               | 1.5                  | 10              | 9.86                | 0.05                   |
| <b>7</b>       | 105              | Lighting         | 4                       | 0.48               | 1.5                  | 10              | 9.50                | 0.01                   |
| <b>8</b>       | 100              | Lighting         | 6                       | 0.45               | 1.5                  | 10              | 9.00                | 0.01                   |
| <b>9</b>       | 660              | Lighting         | 7                       | 3.00               | 1.5                  | 10              | 9.12                | 0.05                   |
| <b>10</b>      | 2100             | Sockets          | 7                       | 9.55               | 2.5                  | 16              | 28.52               | 0.17                   |
| <b>11</b>      | 542              | Lighting         | 8                       | 2.46               | 1.5                  | 10              | 9.74                | 0.04                   |
| <b>12</b>      | 410              | Lighting         | 5                       | 1.86               | 1.5                  | 10              | 12.05               | 0.04                   |
| <b>13</b>      | 330              | Lighting         | 5                       | 1.50               | 1.5                  | 10              | 17.04               | 0.05                   |
| <b>14</b>      | 2100             | Sockets          | 7                       | 9.55               | 2.5                  | 16              | 18.74               | 0.11                   |
| <b>15</b>      | 1200             | Sockets          | 4                       | 5.45               | 2.5                  | 16              | 19.00               | 0.07                   |
| <b>16</b>      | 2500             | Washing Mach.    | 1                       | 11.36              | 2.5                  | 16              | 4.70                | 0.03                   |

**Total Power:20747W**

## Cost Of Project

| Material Item       | Specification                                 | Total Qty | Unit   | Unit Price | Total Price |
|---------------------|-----------------------------------------------|-----------|--------|------------|-------------|
| LED Panel (60x60)   | 36W Integrated LED                            | 2         | Pcs    | 500        | 1000        |
| LED Panel (Surface) | 18W / 16W LED Models                          | 11        | Pcs    | 350        | 3850        |
| LED Mirror Light    | 10W LED                                       | 2         | Pcs    | 2300       | 4600        |
| Waterproof Fixture  | IP44 Rated                                    | 10        | Pcs    | 450        | 4500        |
| Incandescent Lamps  | 150W / 100W / 60W E27 Base                    | 14        | Pcs    | 55         | 770         |
| Power Cable         | 3x2.5 mm <sup>2</sup> NYM (Sockets + Paylar)  | 350       | Meters | 45         | 15750       |
| Lighting Cable      | 3x1.5 mm <sup>2</sup> NYM (Lighting + Paylar) | 300       | Meters | 30         | 9000        |
| Wall Sockets        | Grounded Type                                 | 34        | Pcs    | 140        | 4480        |
| Light Switches      | Single / Commutator / Vavien                  | 18        | Pcs    | 150        | 2700        |
| Circuit Breakers    | 10A, 16A, 25A MCBs                            | 14        | Pcs    | 270        | 3780        |
| Distribution Board  | 18-Way Main Panel                             | 1         | Pcs    | 1.500      | 1500        |
| Installation Kit    | Conduits, Boxes, Accessories                  | 1         | Lot    | 5.000      | 5000        |

Total cost: 56510

## SECOND HOUSE

### Luminous Flux and Power Calculations for Internal and External Illumination

Basement:

$$A = 53.0 \text{ m}^2 \quad \text{target lux: } 50 \quad b = 5.8 \text{ m} \quad l = 9.14 \text{ m} \quad h_k = 2.80 - 0.10 = 2.70$$

$$K = \frac{5.8 \times 8 + 9.14 \times 2}{10 \times 2.70} = 2.4 \quad \eta = 0.49 \quad d = 1.60$$

$$\Phi_{cal} = \frac{1.60 \times 50 \times 53}{0.49} = 9217 \text{ lm} \quad (\text{louvered direct light fitting})$$

for 36W LED lower luminaire (4000lm)

$$n = \frac{9217}{4000} = 2.3 \rightarrow 2 \text{ units}$$

Living room:

$$A = 24.5 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 4.0 \text{ m} \quad l = 6.12 \text{ m} \quad h_k = 2.80 - (0.05 + 0.75) = 2.0 \text{ m}$$

$$K = \frac{4.0 \times 8 + 6.12 \times 2}{10 \times 2.0} = 2.2 \quad \eta = 0.42 \quad d = 1.55$$

$$\Phi_{cal} = \frac{24.5 \times 100 \times 1.55}{0.42} = 9042 \text{ lm} \quad (\text{direct lighting with Plastic diffuser LED})$$

for 60x60cm LED panel 36W (3400lm)

$$n = \frac{9042}{3400} = 2.54 \rightarrow 2 \text{ units}$$

Entrance hall:

$$A = 5.7 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 2.1 \text{ m} \quad l = 2.7 \text{ m} \quad h_k = 2.80 - 0.10 = 2.70 \text{ m}$$

$$K = \frac{2.1 \times 8 + 2.7 \times 2}{10 \times 2.70} = 0.82 \quad \eta = 0.20 \quad d = 1.55$$

$$\Phi_{cal} = \frac{5.7 \times 100 \times 1.55}{0.20} = 4417 \text{ lm} \quad (\text{Semi direct lighting with plastic diffuser led})$$

for 12W opal LED panel: (1100lm)

$$n = \frac{4417}{1100} = 4 \text{ units}$$

Kitchen:

$$A = 12.2 \text{ m}^2 \quad \text{target lux: } 300 \quad b = 3.0 \text{ m} \quad l = 4.07 \text{ m} \quad h_k = 2.80 - 0.85 = 1.95$$

$$K = \frac{3 \times 8 + 4.07 \times 2}{10 \times 1.95} = 1.65 \quad \eta = 0.33 \quad d = 1.70$$

$$\Phi_{cal} = \frac{12.2 \times 300 \times 1.70}{0.33} = 18800 \text{ lm} \quad (\text{louvered semi-indirect linear led lighting})$$

for integrated high output LED module 24W (2400lm)

$$n = \frac{18800}{2400} = 7.5 \rightarrow \text{we use 3 units}$$

"As the area is under 20m<sup>2</sup>, using 3 modules instead of 7 is more appropriate to provide a balanced illumination level without causing visual fatigue."



lower hall:

$$A = 2,0 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 1,0 \text{ m} \quad l = 2,0 \text{ m} \quad h_k = 2,80 - (0,20) = 2,60 \text{ m}$$

$$k = \frac{1,0 \times 8 + 2,0 \times 2}{10 \times 2,60} = 0,46 \quad \eta = 0,29 \quad d = 1,45$$

$$\Phi_{\text{cal}} = \frac{2,0 \times 100 \times 1,45}{0,29} = 1000 \text{ lm} \quad (\text{semi transparent reflective incandescent})$$

for standard clear incandescent 75w

$$n = \frac{1000}{1000} = 1 \text{ unit}$$

Stairs:

$$A = 6,0 \text{ m}^2 \quad \text{target lux: } 150 \quad b = 2,2 \text{ m} \quad l = 2,75 \text{ m} \quad h_k = 2,80 - 0,70 = 2,10$$

$$k = \frac{2,2 \times 8 + 2,75 \times 2}{10 \times 2,10} = 1,03 \quad \eta = 0,29 \quad d = 1,45$$

$$\Phi_{\text{cal}} = \frac{6 \times 150 \times 1,45}{0,29} = 4500 \text{ lm} \quad (\text{semi direct lighting with translucent reflector})$$

for 12w LED module opal diffuser (1080lm)

$$n = \frac{4500}{1080} = 4 \text{ units}$$

Restroom:

$$A = 2,3 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 1,3 \text{ m} \quad l = 1,77 \text{ m} \quad h_k = 2,80 - (0,05 + 0,85) = 1,90 \text{ m}$$

$$k = \frac{1,3 \times 8 + 1,77 \times 2}{10 \times 1,90} = 0,71 \quad \eta = 0,20 \quad d = 1,55$$

$$\Phi_{\text{cal}} = \frac{2,3 \times 100 \times 1,55}{0,20} = 1700 \text{ lm} \quad (\text{Semi indirect lighting with plastic diffuser})$$

for LED downlight with diffuser 9w (850 lm)

$$n = \frac{1700}{850} = 2 \text{ units}$$

Balcony:

$$A = 9,8 \text{ m}^2 \quad \text{target lux: } 50 \quad b = 2,2 \text{ m} \quad l = 4,45 \text{ m} \quad h_k = 2,80 - 0,10 = 2,70$$

$$k = \frac{2,2 \times 8 + 4,45 \times 2}{10 \times 2,70} = 0,9 \quad \eta = 0,29 \quad d = 1,45$$

$$\Phi_{\text{cal}} = \frac{9,8 \times 50 \times 1,45}{0,29} = 2450 \text{ lm} \quad (\text{Semi indirect lighting with translucent reflector})$$

for IP44 glass plate incandescent 100w (1400lm)

$$n = \frac{2450}{1400} = 1,75 \rightarrow 2 \text{ units}$$



### Bathroom (1st floor):

$$A = 5,8 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 2,1 \text{ m} \quad l = 2,76 \text{ m} \quad h_k = 2,80 - 0,85 = 1,95 \text{ m}$$
$$k = \frac{2,1 \times 8 + 2,76 \times 2}{10 \times 1,95} = 1,1 \quad \eta = 0,20 \quad d = 1,55$$

$$\Phi_{\text{cal}} = \frac{5,8 \times 100 \times 1,55}{0,20} = 4400 \text{ lm} \quad (\text{Semi indirect lighting with plastic diffuser})$$

for IP44 LED ceiling light 36w (2000lm)

$$n = \frac{4400}{2000} = 2,2 \rightarrow \text{we will use 1 unit this}$$

for 12w LED for the mirror (1100lm)

$$n = \frac{2400}{1100} = 2,18 \rightarrow \text{we will use 1 unit this}$$

### Upper Hall:

$$A = 4,0 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 1,8 \text{ m} \quad l = 2,22 \text{ m} \quad h_k = 2,80 \text{ m}$$

$$k = \frac{1,8 \times 8 + 2,22 \times 2}{10 \times 2,8} = 0,67 \quad \eta = 0,20 \quad d = 1,55$$

(semi indirect lighting with plastic diffuser)

$$\Phi_{\text{cal}} = \frac{4,0 \times 100 \times 1,55}{0,20} = 3100 \text{ lm}$$

for 18w ceiling light (1400lm) LED

$$n = \frac{3100}{1400} = 2 \text{ units}$$

### Master Bedroom:

$$A = 17,6 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 3,8 \text{ m} \quad l = 4,63 \text{ m} \quad h_k = 2,80 - (0,20 + 0,75) = 1,85 \text{ m}$$

$$k = \frac{3,8 \times 8 + 4,63 \times 2}{10 \times 1,85} = 2,14 \quad \eta = 0,35 \quad d = 1,55$$

$$\Phi_{\text{cal}} = \frac{17,6 \times 100 \times 1,55}{0,35} = 7780 \text{ lm} \quad (\text{Semi indirect lighting with plastic diffuser})$$

for 18w LED ceiling light (1400lm)

$$n = \frac{7780}{1400} = 5,5 \rightarrow \text{we use 2 units this}$$

for 5w LED reading light (500lm)

$$n = 2 \text{ units}$$

### Closet room:

$$A = 6,2 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 2 \text{ m} \quad l = 3,1 \text{ m} \quad h_L = 2,80 - 0,85 = 1,95 \text{ m}$$

$$K = \frac{2 \times 8 + 3,1 \times 2}{10 \times 1,95} = 1,14 \quad \eta = 0,23 \quad d = 1,55$$

$$\Phi_{\text{cal}} = \frac{6,2 \times 100 \times 1,55}{0,23} = 4178 \text{ lm} \quad (\text{Semi indirect lighting with plastic diffuser})$$

for LED downlight with diffuser 5w (500lm)

$$n = \frac{4178}{500} = 8,0 \rightarrow \text{we will use 6 units}$$

### Master Bathroom:

$$A = 4,2 \text{ m}^2 \quad \text{target lux: } 100 \quad b = 1,8 \text{ m} \quad l = 2,33 \text{ m} \quad h_L = 2,80 - 0,85 = 1,95 \text{ m}$$

$$K = \frac{1,8 \times 8 + 2,33 \times 2}{10 \times 1,95} = 0,98 \quad \eta = 0,20 \quad d = 1,55$$

(Semi indirect lighting with plastic diffuser)

$$\Phi_{\text{cal}} = \frac{4,2 \times 100 \times 1,55}{0,20} = 3000 \text{ lm}$$

for IP44 LED ceiling light 24w (2400lm)

$$n = \frac{3000}{2400} = 1,25 \rightarrow 1 \text{ unit}$$

7w LED for the mirror (700lm)

$$n = \frac{600}{700} \approx 1 \text{ unit}$$

### Stairs 2:

$$A = 4,25 \text{ m}^2 \quad \text{target lux: } 50 \quad b = 1,0 \text{ m} \quad l = 4,25 \text{ m} \quad h_L = 2,80 - 0,10 = 2,70 \text{ m}$$

$$K = \frac{1 \times 8 + 4,25 \times 2}{10 \times 2,7} = 0,6 \quad \eta = 0,29 \quad d = 1,45$$

(Semi indirect lighting with transducer reflector)

$$\Phi_{\text{cal}} = \frac{4,25 \times 50 \times 1,45}{0,29} = 1062 \text{ lm}$$

for IP44 glass globe 60w incandescent with sensor

$$n = \frac{1064}{800} = 1,3 \rightarrow 1 \text{ unit}$$

### Room 1:

$$A = 11,1 \text{ m}^2 \quad \text{target lux: } 75 \quad b = 3 \text{ m} \quad l = 3,7 \text{ m} \quad h_L = 2,80 - (0,20 + 0,75) = 1,85 \text{ m}$$

$$K = \frac{3 \times 8 + 3,7 \times 2}{10 \times 1,85} = 1,7 \quad \eta = 0,28 \quad d = 1,55$$

(Semi indirect lighting with plastic diffuser)

$$\Phi_{\text{cal}} = \frac{11,1 \times 75 \times 1,55}{0,28} = 5600 \text{ lm}$$

for 24w LED ceiling light (2400lm)

$$n = \frac{5600}{2400} = 2 \rightarrow \text{but it's too much for 12 m}^2 \text{ bedroom so we will use 1 unit this}$$

### Room 1:

$$A = 12,1 \quad \text{target lux: } 75 \quad b = 3 \text{ m} \quad l = 4,03 \text{ m} \quad h_L = 2,80 - (0,20 + 0,75) = 1,85 \text{ m}$$

$$K = \frac{3 \times 8 + 4,03 \times 2}{10 \times 1,85} = 1,73 \quad \eta = 0,28 \quad d = 1,55$$

$$\Phi_{\text{cal}} = \frac{12,1 \times 75 \times 1,55}{0,28} = 5000 \text{ lm} \quad (\text{Semi indirect lighting with plastic diffuser})$$

for 24w LED ceiling light (2400lm)

$$n = \frac{5000}{2400} = 2 \rightarrow \text{we will use 1 unit}$$



| Room / Area          | Area (m²) | Target Lux (lx) | Calculated Lumens (lm) | Fixture / Lamp Selection                               |
|----------------------|-----------|-----------------|------------------------|--------------------------------------------------------|
| Living Room          | 24.5      | 100             | 9042                   | 2 units 60x60 LED panel 36W                            |
| Entrance Hall        | 5.7       | 100             | 4417                   | 4 units 12W opal LED panel                             |
| Stairs               | 6.0       | 150             | 4500                   | 4 Units 12W Led module opal diffuser                   |
| Lower Hall           | 2.0       | 100             | 1000                   | Standard clear incandescent 75W                        |
| Restroom             | 2.3       | 100             | 1700                   | 2 units LED downlight with diffuser 9W                 |
| Kitchen              | 12.2      | 300             | 18800                  | 3 units integrated high output LED module 24W          |
| Basement             | 53.0      | 50              | 9217                   | 2 units 36W LED louvered luminaire                     |
| Balcony (1st Floor)  | 9.8       | 50              | 2450                   | 2 unit IP44 Glass Globe 100W Incandescent              |
| Room 1               | 12.1      | 75              | 5000                   | LED ceiling light 24 W                                 |
| Room 2               | 11.1      | 75              | 4600                   | LED ceiling light 24 W                                 |
| Bathroom (1st Floor) | 5.8       | 100             | 4400                   | IP44 LED ceiling light 36 W and 12W LED                |
| Upper Hall           | 4.0       | 100             | 3100                   | 2 units 18W ceiling light                              |
| Master Bedroom       | 17.6      | 100             | 7780                   | 2 units 18W ceiling light and 2 units 5W reading light |
| Closet Room          | 6.2       | 100             | 4178                   | 6 units downlight with diffuser 5W                     |
| Master Bathroom      | 4.2       | 100             | 3000                   | IP44 LED ceiling light 24 W and 7W LED                 |
| Stairs 2             | 4.25      | 50              | 1062                   | 60W IP44 Glass Globe incandescent with moving sensor   |

## Light Sources and Luminaires

| Code | Luminaire Type                                 | Lamp Type    | Power (W)   | Luminous Flux (lm) | IP Rating |
|------|------------------------------------------------|--------------|-------------|--------------------|-----------|
| A1   | 60x60 LED Panel (2 units)                      | LED          | 36 W        | 3200 lm            | IP20      |
| A2   | Opal LED Panel (4 units)                       | LED          | 12 W        | 1000 lm            | IP20      |
| A3   | LED Module Opal Diffuser (4 units)             | LED Module   | 12 W        | 1050 lm            | IP20      |
| B1   | Standard Clear Incandescent (1 units)          | Incandescent | 75 W        | 900 lm             | IP20      |
| A4   | LED Downlight with Diffuser (2 units)          | LED          | 9 W         | 760 lm             | IP20      |
| A5   | Integrated High Output LED Module (3 units)    | LED Module   | 24 W        | 2400 lm            | IP20      |
| A6   | LED Louvered Luminaire (2 units)               | LED          | 36 W        | 3400 lm            | IP20      |
| B2   | Glass Globe Luminaire (2 units)                | Incandescent | 100 W       | 1300 lm            | IP44      |
| A7   | LED Ceiling Light (1 unit)                     | LED          | 24 W        | 2000 lm            | IP20      |
| A8   | LED Ceiling Light (1 unit)                     | LED          | 24 W        | 2000 lm            | IP20      |
| A9   | Combo: LED Ceiling Light & LED (1 unit)        | LED          | 36 W + 12 W | 3200 + 1000 lm     | IP44      |
| A10  | Ceiling Light (2 units)                        | LED          | 18 W        | 1500 lm            | IP20      |
| A11  | Combo: Ceiling Light & Reading Light (2 units) | LED          | 18 W + 5 W  | 1500 + 400 lm      | IP20      |
| A12  | Downlight with Diffuser (6 units)              | LED          | 5 W         | 400 lm             | IP20      |
| A13  | Combo: LED Ceiling Light & LED (1 unit)        | LED          | 24 W + 7 W  | 2000 + 600 lm      | IP44      |
| B3   | Glass Globe with Motion Sensor (1 unit)        | Incandescent | 60 W        | 710 lm             | IP44      |

## Loading Table

| Line No | Power | Load Type     | Current (A) | Section | Fuse (A) | Distance (m) | Volt. Drop |
|---------|-------|---------------|-------------|---------|----------|--------------|------------|
| 1       | 72    | Lighting      | 0.33        | 1.5     | 10       | 8.50         | 0.01       |
| 2       | 1200  | Sockets       | 5.45        | 2.5     | 16       | 25.00        | 0.23       |
| 3       | 2000  | Oven          | 9.09        | 2.5     | 16       | 8.10         | 0.11       |
| 4       | 2000  | Kettle        | 9.09        | 2.5     | 16       | 6.52         | 0.09       |
| 5       | 2500  | Dishwasher    | 11.36       | 2.5     | 16       | 4.08         | 0.07       |
| 6       | 2200  | Sockets       | 10.00       | 2.5     | 16       | 13.87        | 0.23       |
| 7       | 165   | Lighting      | 0.75        | 1.5     | 10       | 6.44         | 0.01       |
| 8       | 96    | Lighting      | 0.44        | 1.5     | 10       | 9.70         | 0.01       |
| 9       | 1800  | Sockets       | 8.18        | 2.5     | 16       | 12.76        | 0.17       |
| 10      | 132   | Lighting      | 0.60        | 1.5     | 10       | 9.34         | 0.01       |
| 11      | 61    | Lighting      | 0.28        | 1.5     | 10       | 8.04         | 0.00       |
| 12      | 46    | Lighting      | 0.21        | 1.5     | 10       | 6.15         | 0.00       |
| 13      | 2100  | Sockets       | 9.55        | 2.5     | 16       | 18.54        | 0.29       |
| 14      | 2500  | Washing Mach. | 11.36       | 2.5     | 16       | 4.50         | 0.08       |
| 15      | 172   | Lighting      | 0.78        | 1.5     | 10       | 12.08        | 0.02       |
| 16      | 2100  | Sockets       | 9.55        | 2.5     | 16       | 28.52        | 0.45       |
| 17      | 160   | Lighting      | 0.73        | 1.5     | 10       | 10.73        | 0.02       |



Total Power: 19304

## Cost Of Project

| Material Item                        | Specification                                              | Total Qty | Unit   | Unit Price | Total Price |
|--------------------------------------|------------------------------------------------------------|-----------|--------|------------|-------------|
| 60x60 LED Panel                      | 36W, Recessed Square Panel                                 | 2         | Pcs    | 500        | 1000        |
| Opal LED Panel                       | 12W, Slim Diffused Panel                                   | 4         | Pcs    | 350        | 1400        |
| LED Module with Opal Diffuser        | 12W, High Efficiency Module                                | 4         | Pcs    | 1000       | 4000        |
| Standard Clear Incandescent          | 75W, E27 Filament Lamp                                     | 1         | Pcs    | 650        | 650         |
| LED Downlight with Diffuser          | 9W, Recessed Spot                                          | 2         | Pcs    | 1400       | 2800        |
| Integrated High Output LED Module    | 24W, Linear Integrated Module                              | 3         | Pcs    | 500        | 1500        |
| LED Louvered Luminaire               | 36W, Anti-Glare Office Type                                | 2         | Pcs    | 2400       | 4800        |
| Glass Globe Incandescent             | 100W, IP44 Moisture Proof Waterproof Base                  | 2         | Pcs    | 450        | 900         |
| LED Ceiling Light                    | 24W, Surface Mounted                                       | 2         | Pcs    | 460        | 920         |
| Combo: LED Ceiling Light + LED       | 36W + 12W Dual Driver Configuration, IP44                  | 1         | Pcs    | 1600       | 1600        |
| Ceiling Light                        | 18W, Circular Surface Mounted                              | 2         | Pcs    | 250        | 500         |
| Combo: Ceiling Light + Reading Light | 18W Ambient + 5W Focused Reading Light                     | 2         | Pcs    | 600        | 1200        |
| Downlight with Diffuser              | 5W, Accent Lighting Mini Spot                              | 6         | Pcs    | 350        | 2100        |
| Combo: LED Ceiling Light + LED       | 24W + 7W Multi-Purpose Fixture, IP44                       | 1         | Pcs    | 500        | 500         |
| Glass Globe with Motion Sensor       | 60W, IP44 Infrared Sensor Integral                         | 1         | Pcs    | 600        | 600         |
| Power Cable                          | 3x2.5 mm <sup>2</sup> NYM (Cu/PVC/PVC) Heavy Duty          | 370       | Meters | 45         | 16650       |
| Lighting Cable                       | 3x1.5 mm <sup>2</sup> NYM (Cu/PVC/PVC) Standard            | 350       | Meters | 30         | 10500       |
| Light Switches                       | Single/Commutator/Vavien                                   | 17        | Pcs    | 320        | 5440        |
| Distribution Board                   | 18 Way Main Panel                                          | 1         | Pcs    | 1500       | 1500        |
| Installation Board                   | Conduits, Boxes, Accessories                               | 1         | Lot    | 5000       | 5000        |
| Electrical                           | 16A, 250V Flush Mounted Earth Socket with Child Protection | 33        | Pcs    | 140        | 4620        |

Total Price: 68180

## Video of Project

<https://youtu.be/eYsRW0s9ccM>

## Conclusion

This project presented two alternative residential lighting system designs based on illumination engineering

principles. For each design, the required illumination levels were determined through luminous flux and lux calculations, and appropriate luminaires were selected according to the characteristics of each space. Electrical loading tables, cable sizing, voltage drop calculations, and project cost estimations were also completed to evaluate the technical feasibility of the installations.

The comparison of the two alternatives demonstrated how different lighting technologies influence both total installed power and overall project cost. While the first design combines LED and incandescent lighting solutions, the second design primarily utilizes LED-based luminaires to improve energy efficiency while maintaining the required lighting performance.

Overall, this project provided practical experience in residential lighting design by integrating illumination planning, electrical installation calculations, and economic evaluation into a complete engineering design process.